MIMIC

MIMIC-IV / Hosp / admissions table

The *admissions* table gives information regarding a patient's admission to the hospital. Since each unique hospital visit for a patient is assigned a unique `hadm_id`, the *admissions* table can be considered as a definition table for `hadm_id`. Information available includes timing information for admission and discharge, demographic information, the source of the admission, and so on.

Links to

- *patients* on `subject_id`

Important considerations

- The data is sourced from the admission, discharge and transfer database from the hospital (often referred to as 'ADT' data).
- Organ donor accounts are sometimes created for patients who died in the hospital. These are distinct hospital admissions with very short, sometimes negative lengths of stay. Furthermore, their `deathtime` is frequently the same as the earlier patient admission's `deathtime`.

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER NOT NULL
`admittime`	TIMESTAMP NOT NULL
`dischtime`	TIMESTAMP
`deathtime`	TIMESTAMP
`admission_type`	VARCHAR(40) NOT NULL

admit_provider_id	VARCHAR(10)
admission_location	VARCHAR(60)
discharge_location	VARCHAR(60)
insurance	VARCHAR(255)
language	VARCHAR(10)
marital_status	VARCHAR(30)
race	VARCHAR(80)
edregtime	TIMESTAMP
edouttime	TIMESTAMP
hospital_expire_flag	SMALLINT

Detailed description

The *admissions* table defines all hospitalizations in the database. Hospitalizations are assigned a unique random integer known as the `hadm_id`.

`subject_id`, `hadm_id`

Each row of this table contains a unique `hadm_id`, which represents a single patient's admission to the hospital. `hadm_id` ranges from 2000000 - 2999999. It is possible for this table to have duplicate `subject_id`, indicating that a single patient had multiple admissions to the hospital. The ADMISSIONS table can be linked to the PATIENTS table using `subject_id`.

`admittime`, `dischtime`, `deathtime`

`admittime` provides the date and time the patient was admitted to the hospital, while `dischtime` provides the date and time the patient was discharged from the hospital. If applicable, `deathtime` provides the time of in-hospital death for the patient. Note that `deathtime` is only present if the patient died in-hospital, and is almost always the same as the patient's `dischtime`. However, there can be some discrepancies due to typographical errors.

`admission_type`

`admission_type` is useful for classifying the urgency of the admission. There are 9 possibilities: 'AMBULATORY OBSERVATION', 'DIRECT EMER.', 'DIRECT OBSERVATION', 'ELECTIVE', 'EU OBSERVATION', 'EW EMER.', 'OBSERVATION ADMIT', 'SURGICAL SAME DAY ADMISSION', 'URGENT'.

`admit_provider_id`

`admit_provider_id` provides an anonymous identifier for the provider who admitted the patient. Provider identifiers follow a consistent pattern: the letter "P", followed by either three numbers, followed by two letters or two numbers. For example, "P003AB", "P00102", "P1248B", etc. Provider identifiers are randomly generated and do not have any inherent meaning aside from uniquely identifying the same provider across the database.

`admission_location`, `discharge_location`

`admission_location` provides information about the location of the patient prior to arriving at the hospital. Note that as the emergency room is technically a clinic, patients who are admitted via the emergency room usually have it as their admission location.

Similarly, `discharge_location` is the disposition of the patient after they are discharged from the hospital.

ASSOCIATION WITH UB-04 BILLING CODES

`admission_location` and `discharge_location` are associated with internal hospital `ibax` codes which aren't provided in MIMIC-IV. These internal codes tend to align with UB-04 billing codes.

In some cases more than one internal code is associated with a given `admission_location` and `discharge_location`. This can either be do to; 1) multiple codes being used by the hospital for the same `admission_location` or `discharge_location`, or 2) during de-identification multiple internal codes may be combined into a single `admission_location` or `discharge_location`.

In the tables below, we provide the matching UB-04 code(s) for the most common `ibax` codes for a given `admission_location` and `discharge_location`, when applicable. In cases where more than one code is given, if this combination is due to 1) in the above paragraph, the additional code must have at least 10% of the entires of the most

common code.

Admission UB-04 mappings: admission_location | UB-04 code(s) – | – PHYSICIAN REFERRAL | 1, 3 WALK-IN/SELF REFERRAL | 1 AMBULATORY SURGERY TRANSFER | 1, 2, 6 INFORMATION NOT AVAILABLE | 1, 9 CLINIC REFERRAL | 2, 8 PROCEDURE SITE | 2 PACU | 2 TRANSFER FROM HOSPITAL | 4, 6 TRANSFER FROM SKILLED NURSING FACILITY | 5 EMERGENCY ROOM | 1, 2, 7 INTERNAL TRANSFER TO OR FROM PSYCH | none

Discharge UB-04 mappings: discharge_location | UB-04 code(s) – | – HOME | 01 ACUTE HOSPITAL | 02, 81, 86 SKILLED NURSING FACILITY | 03, 64 ASSISTED LIVING | 04 HEALTHCARE FACILITY | 05, 43 HOME HEALTH CARE | 06 AGAINST ADVICE | 07 DIED | 20 OTHER FACILITY | 21, 70 HOSPICE | 50, 51 REHAB | 62 CHRONIC/LONG TERM ACUTE CARE | 63 PSYCH FACILITY | 65 OTHER FACILITY | 70

UB-04 documentation online often provides more detail than found in the `admission_location` and `discharge_location` text, particularly for discharges.

`insurance`, `language`, `marital_status`, `ethnicity`

The `insurance`, `language`, `marital_status`, and `ethnicity` columns provide information about patient demographics for the given hospitalization. Note that as this data is documented for each hospital admission, they may change from stay to stay.

`edregtime`, `edouttime`

The date and time at which the patient was registered and discharged from the emergency department.

`hospital_expire_flag`

This is a binary flag which indicates whether the patient died within the given hospitalization. `1` indicates death in the hospital, and `0` indicates survival to hospital discharge.

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MIMIC

MIMIC-IV / Hosp / d_hcpcs

The *d_hcpcs* table

The *d_hcpcs* table is used to acquire human readable definitions for the codes used in the *hcpcsevents* table. The concepts primarily correspond to hospital billing, and are mostly CPT codes. Unfortunately due to licensing restrictions not all code definitions are available.

Links to

- *hcpcsevents* on `code`

Table columns

Name	Postgres data type
`code`	CHAR(5) NOT NULL
`category`	SMALLINT
`long_description`	TEXT
`short_description`	VARCHAR(180)

Detailed description

`code`
A five character code which uniquely represents the event.

`category`
Broad classification of the code.

`long_description`, `short_description`

Textual descriptions of the `code` listed for the given row.

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MIMIC

MIMIC-IV / Hosp / d_icd_diagnoses

The *d_icd_diagnoses* table

This table defines International Classification of Diseases (ICD) Version 9 and 10 codes for **diagnoses**. These codes are assigned at the end of the patient's stay and are used by the hospital to bill for care provided.

Links to

- *diagnoses_icd* ON `icd_code` and `icd_version`

Table columns

Name	Postgres data type
`icd_code`	CHAR(7) NOT NULL
`icd_version`	INTEGER NOT NULL
`long_title`	VARCHAR(255)

Detailed Description

`icd_code`, `icd_version`
`icd_code` is the International Coding Definitions (ICD) code.

There are two versions for this coding system: version 9 (ICD-9) and version 10 (ICD-10). These can be differentiated using the `icd_version` column. In general, ICD-10 codes are more detailed, though code mappings (or "cross-walks") exist which convert ICD-9 codes to ICD-10 codes.

Both ICD-9 and ICD-10 codes are often presented with a decimal. This decimal is not required for interpretation of an ICD code; i.e. the `icd_code` of '0010' is equivalent to

'001.0'.

ICD-9 and ICD-10 codes have distinct formats: ICD-9 codes are 5 character long strings which are entirely numeric (with the exception of codes prefixed with "E" or "V" which are used for external causes of injury or supplemental classification). Importantly, ICD-9 codes are retained as strings in the database as the leading 0s in codes are meaningful.

ICD-10 codes are 3-7 characters long and always prefixed by a letter followed by a set of numeric values.

`long_title`

The `long_title` provides the meaning of the ICD code. For example, the ICD-9 code 0010 has `long_title` "Cholera due to vibrio cholerae".

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MIMIC

MIMIC-IV / Hosp / d_icd_procedures

The *d_icd_procedures* table

This table defines International Classification of Diseases (ICD) codes for **procedures**. These codes are assigned at the end of the patient's stay and are used by the hospital to bill for care provided. They can further be used to identify if certain procedures have been performed (e.g. surgery).

Links to

- *procedures_icd* on `icd_code` and `icd_version`

Brief summary

Table columns

Name	Postgres data type
`icd_code`	CHAR(7) NOT NULL
`icd_version`	INTEGER NOT NULL
`long_title`	VARCHAR(255)

Detailed Description

`icd_code`, `icd_version`
`icd_code` is the International Coding Definitions (ICD) code.

There are two versions for this coding system: version 9 (ICD-9) and version 10 (ICD-10). These can be differentiated using the `icd_version` column. In general, ICD-10 codes are more detailed, though code mappings (or "cross-walks") exist which convert

ICD-9 codes to ICD-10 codes.

Both ICD-9 and ICD-10 codes are often presented with a decimal. This decimal is not required for interpretation of an ICD code; i.e. the `icd_code` of '0010' is equivalent to '001.0'.

`long_title`

The title fields provide a brief definition for the given procedure code in ``.

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MIMIC

MIMIC-IV / Hosp / d_labitems

d_labitems

d_labitems contains definitions for all `itemid` associated with lab measurements in the MIMIC database. All data in *labevents* link to the *d_labitems* table. Each unique (`fluid`, `category`, `label`) tuple in the hospital database was assigned an `itemid` in this table, and the use of this `itemid` facilitates efficient storage and querying of the data.

Laboratory data contains information collected and recorded in the hospital laboratory database. This includes measurements made in wards within the hospital and clinics outside the hospital. Most concepts in this table have been mapped to LOINC codes, an openly available ontology which facilitates interoperability.

Links to

- *labevents* on `itemid`

Important considerations

This table used to contain a column called `loinc_code`, which stored standardized identifiers for laboratory measurements. To support ongoing improvement of these labels, the assignment of LOINC codes is now done in the MIMIC Code Repository.

Table columns

Name	Postgres data type
`itemid`	INTEGER
`label`	VARCHAR(50)
`fluid`	VARCHAR(50)
`category`	VARCHAR(50)

Detailed Description

`itemid`

A unique identifier for a laboratory concept. `itemid` is unique to each row, and can be used to identify data in labevents associated with a specific concept.

`label`

The `label` column describes the concept which is represented by the `itemid`.

`fluid`

`fluid` describes the substance on which the measurement was made. For example, chemistry measurements are frequently performed on blood, which is listed in this column as 'BLOOD'. Many of these measurements are also acquirable on other fluids, such as urine, and this column differentiates these distinct concepts.

`category`

`category` provides higher level information as to the type of measurement. For example, a category of 'ABG' indicates that the measurement is an arterial blood gas.

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MIMIC

MIMIC-IV / Hosp / diagnoses_icd

diagnoses_icd

During routine hospital care, patients are billed by the *hospital* for diagnoses associated with their hospital stay. This table contains a record of all diagnoses a patient was billed for during their hospital stay using the ICD-9 and ICD-10 ontologies. Diagnoses are billed on hospital discharge, and are determined by trained persons who read signed clinical notes.

Links to

- *d_icd_diagnoses* ON `icd_code` and `icd_version`

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER NOT NULL
`seq_num`	INTEGER NOT NULL
`icd_code`	VARCHAR(7)
`icd_version`	INTEGER

Detailed Description

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`seq_num`

The priority assigned to the diagnoses. The priority can be interpreted as a ranking of which diagnoses are "important", but many caveats to this broad statement exist. For example, patients who are diagnosed with sepsis must have sepsis as their *2nd* billed condition. The 1st billed condition must be the infectious agent. There's also less importance placed on ranking low priority diagnoses "correctly" (as there may be no correct ordering of the priority of the 5th - 10th diagnosis codes, for example).

`icd_code`, `icd_version`

`icd_code` is the International Coding Definitions (ICD) code.

There are two versions for this coding system: version 9 (ICD-9) and version 10 (ICD-10). These can be differentiated using the `icd_version` column. In general, ICD-10 codes are more detailed, though code mappings (or "cross-walks") exist which convert ICD-9 codes to ICD-10 codes.

Both ICD-9 and ICD-10 codes are often presented with a decimal. This decimal is not required for interpretation of an ICD code; i.e. the `icd_code` of '0010' is equivalent to '001.0'.

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MIMIC

MIMIC-IV / Hosp / drgcodes

drgcodes

Diagnosis related groups (DRGs) are used by the hospital to obtain reimbursement for a patient's hospital stay. The codes correspond to the primary reason for a patient's stay at the hospital.

Table columns

Name	Postgres data type
`subject_id`	INTEGER
`hadm_id`	INTEGER
`drg_type`	VARCHAR(4)
`drg_code`	VARCHAR(10)
`description`	VARCHAR(195)
`drg_severity`	SMALLINT
`drg_mortality`	SMALLINT

Detailed Description

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`drg_type`

The specific DRG ontology used for the code.

`drg_code`

The DRG code.

`description`

A description for the given DRG code.

`drg_severity`, `drg_mortality`

Some DRG ontologies further qualify the patient severity of illness and likelihood of mortality, which are recorded here.

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MIMIC

MIMIC-IV / Hosp / emar

emar

The EMAR table is used to record administrations of a given medicine to an individual patient. Records in this table are populated by bedside nursing staff scanning barcodes associated with the medicine and the patient.

Links to

- *emar_detail* on `emar_id`
- *pharmacy* on `pharmacy_id`
- *prescriptions* on `pharmacy_id`
- *poe* on `poe_id`

Important considerations

- The eMAR system was implemented during 2011-2013. As a result, eMAR data is not available for all patients.

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER
`emar_id`	VARCHAR(25) NOT NULL
`emar_seq`	INTEGER NOT NULL
`poe_id`	VARCHAR(25) NOT NULL
`pharmacy_id`	INTEGER

enter_provider_id	VARCHAR(10)
charttime	TIMESTAMP NOT NULL
medication	TEXT
event_txt	VARCHAR(100)
scheduletime	TIMESTAMP
storetime	TIMESTAMP NOT NULL

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`emar_id`, `emar_seq`

Identifiers for the eMAR table. `emar_id` is a unique identifier for each order made in eMAR. `emar_seq` is a consecutive integer which numbers eMAR orders chronologically. `emar_id` is composed of `subject_id` and `emar_seq` in the following pattern: '`subject_id`-`emar_seq`'.

`poe_id`

An identifier which links administrations in *emar* to orders in *poe* and *prescriptions*.

`pharmacy_id`

An identifier which links administrations in *emar* to pharmacy information in the *pharmacy* table.

`enter_provider_id`

`enter_provider_id` provides an anonymous identifier for the provider who entered the information into the eMAR system. Provider identifiers follow a consistent pattern: the letter "P", followed by either three numbers, followed by two letters or two numbers. For

example, “P003AB”, “P00102”, “P1248B”, etc. Provider identifiers are randomly generated and do not have any inherent meaning aside from uniquely identifying the same provider across the database.

`charttime`

The time at which the medication was administered.

`medication`

The name of the medication which was administered.

`event_txt`

Information about the administration. Most frequently `event_txt` is ‘Administered’, but other possible values are ‘Applied’, ‘Confirmed’, ‘Delayed’, ‘Not Given’, and so on.

`scheduletime`

If present, the time at which the administration was scheduled.

`storetime`

The time at which the administration was documented in the eMAR table.

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MIMIC

MIMIC-IV / Hosp / emar_detail

emar_detail

The *emar_detail* table contains information for each medicine administration made in the EMAR table. Information includes the associated pharmacy order, the dose due, the dose given, and many other parameters associated with the medical administration.

Links to

- *emar* on `emar_id`
- *pharmacy* on `pharmacy_id`

Important considerations

- The eMAR system was implemented during 2011-2013. As a result, eMAR data is not available for all patients.

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`emar_id`	VARCHAR(25) NOT NULL
`emar_seq`	INTEGER NOT NULL
`parent_field_ordinal`	VARCHAR(10)
`administration_type`	VARCHAR(50)
`pharmacy_id`	INTEGER
`barcode_type`	VARCHAR(4)

reason_for_no_barcode	TEXT
complete_dose_not_given	VARCHAR(5)
dose_due	VARCHAR(100)
dose_due_unit	VARCHAR(50)
dose_given	VARCHAR(255)
dose_given_unit	VARCHAR(50)
will_remainder_of_dose_be_given	VARCHAR(5)
product_amount_given	VARCHAR(30)
product_unit	VARCHAR(30)
product_code	VARCHAR(30)
product_description	VARCHAR(255)
product_description_other	VARCHAR(255)
prior_infusion_rate	VARCHAR(40)
infusion_rate	VARCHAR(40)
infusion_rate_adjustment	VARCHAR(50)
infusion_rate_adjustment_amount	VARCHAR(30)
infusion_rate_unit	VARCHAR(30)
route	VARCHAR(10)
infusion_complete	VARCHAR(1)
completion_interval	VARCHAR(50)
new_iv_bag_hung	VARCHAR(1)
continued_infusion_in_other_location	VARCHAR(1)

`restart_interval`	TEXT
`side`	VARCHAR(10)
`site`	VARCHAR(255)
`non_formulary_visual_verification`	VARCHAR(1)

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`emar_id`, `emar_seq`

Identifiers for the eMAR table. `emar_id` is a unique identifier for each order made in eMAR. `emar_seq` is a consecutive integer which numbers eMAR orders chronologically. `emar_id` is composed of `subject_id` and `emar_seq` in the following pattern: '`subject_id`-`emar_seq`'.

`parent_field_ordinal`

`parent_field_ordinal` delineates multiple administrations for the same eMar event, e.g. multiple formulary doses for the full dose. As eMAR requires the administrating provider to scan a barcode for *each* formulary provided to the patient, it is often the case that multiple rows in *emar_detail* correspond to a single row in *emar* (e.g. multiple pills are administered which add up to the desired dose). The structure for *emar_detail* rows is as follows:

- There is one row per eMAR order with a NULL `parent_field_ordinal`: this row usually contains the desired dose for the administration.
- Afterward, if there are N formulary doses, `parent_field_ordinal` will take values '1.1', '1.2', ..., '1.N'.

The most common case occurs when there is only one formulary dose per medication. In this case the `emar_id` will have two rows in the *emar_detail* table: one with a NULL value for `parent_field_ordinal` (usually providing the dose due), and one row with a value of '1.1' for `parent_field_ordinal` (usually providing the actual dose administered).

`administration_type`

The type of administration, including 'IV Bolus', 'IV Infusion', 'Medication Infusion', 'Transdermal Patch', and so on.

`pharmacy_id`
An identifier which allows linking the eMAR order to pharmacy information provided in the *pharmacy* table. Note: rarely the same `emar_id` may have multiple distinct `pharmacy_id` across rows in the *emar_detail* table.

Remaining columns

The remaining columns provide information about the delivery of the formulary dose of the administered medication.

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MIMIC

MIMIC-IV / Hosp / hpcsevents

hcpcsevents

Links to

- *d_hcpcs* on hcpcs_cd

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER NOT NULL
`chartdate`	DATE
`hcpcs_cd`	CHAR(5) NOT NULL
`seq_num`	INTEGER NOT NULL
`short_description`	VARCHAR(180)

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`chartdate`

The date associated with the coded event.

`hcpcs_cd`

A five character code which uniquely represents the event. Link this to `code` in *d_hcpcs* for a longer description of the code.

`seq_num`

An assigned order to HCPCS codes for an individual hospitalization. This order sometimes conveys meaning, e.g. sometimes higher priority, but this is not guaranteed across all codes.

`short_description`

A short textual descriptions of the `hcpcs_cd` listed for the given row.

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MIMIC

MIMIC-IV / Hosp / labevents

labevents

The *labevents* table stores the results of all laboratory measurements made for a single patient. These include hematology measurements, blood gases, chemistry panels, and less common tests such as genetic assays.

Links to

- *d_labitems* on `itemid`

Important considerations

`hadm_id` is assigned to labs close to the hospital stay using the *transfers* table. However this does not always perfectly capture labs proximal to the hospital stay. To be specific, as of v2.1, it is possible to assign 59,327,830 observations with an `hadm_id` using a join to *admissions* with `subject_id`, `admittime`, and `dischtime`. However, only 58,131,956 (98%) of these observations have an `hadm_id` actually stored in the *labevents* table. Users wishing to ensure capture of labs proximal to hospital stays should be aware of this, and use joins with time as necessary.

The following conditions must occur in order for this data to not have an `hadm_id` in the table:

- the lab observation must fall outside of a row in *transfers*
- the lab observation must not have an associated ED stay identifier

Table columns

Name	Postgres data type
`labevent_id`	INTEGER NOT NULL
`subject_id`	INTEGER NOT NULL

hadm_id	INTEGER
specimen_id	INTEGER NOT NULL
itemid	INTEGER NOT NULL
order_provider_id	VARCHAR(10)
charttime	TIMESTAMP(0)
storetime	TIMESTAMP(0)
value	VARCHAR(200)
valuenum	DOUBLE PRECISION
valueuom	VARCHAR(20)
ref_range_lower	DOUBLE PRECISION
ref_range_upper	DOUBLE PRECISION
flag	VARCHAR(10)
priority	VARCHAR(7)
comments	TEXT

`labevent_id`

An integer which is unique for every row in the table.

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`specimen_id`

Uniquely denoted the specimen from which the lab measurement was made. Most lab measurements are made on patient derived samples (specimens) such as blood, urine, and so on. Often multiple measurements are made on the same sample. The `specimen_id` will group measurements made on the same sample, e.g. blood gas measurements made on the same sample of blood.

`itemid`

An identifier which uniquely denotes laboratory concepts.

`order_provider_id`

`order_provider_id` provides an anonymous identifier for the provider who ordered the laboratory measurement. Provider identifiers follow a consistent pattern: the letter "P", followed by either three numbers, followed by two letters or two numbers. For example, "P003AB", "P00102", "P1248B", etc. Provider identifiers are randomly generated and do not have any inherent meaning aside from uniquely identifying the same provider across the database.

`charttime`

The time at which the laboratory measurement was charted. This is usually the time at which the specimen was acquired, and is usually significantly **earlier** than the time at which the measurement is available.

`storetime`

The time at which the measurement was made available in the laboratory system. This is when the information would have been available to care providers.

`value`, `valuenum`

The result of the laboratory measurement and, if it is numeric, the value cast as a numeric data type.

`valueuom`

The unit of measurement for the laboratory concept.

`ref_range_lower`, `ref_range_upper`

Upper and lower reference ranges indicating the normal range for the laboratory

measurements. Values outside the reference ranges are considered abnormal.

`flag`

A brief string mainly used to indicate if the laboratory measurement is abnormal.

`priority`

The priority of the laboratory measurement: either routine or stat (urgent).

`comments`

Deidentified free-text comments associated with the laboratory measurement. Usually these provide information about the sample, whether any notifications were made to care providers regarding the results, considerations for interpretation, or in some cases the comments contain the result of the laboratory itself. Comments which have been fully deidentified (i.e. no information content retained) are present as three underscores: `___`. A `NULL` comment indicates no comment was made for the row.

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MIMIC

MIMIC-IV / Hosp / microbiologyevents

microbiologyevents

Microbiology tests are a common procedure to check for infectious growth and to assess which antibiotic treatments are most effective.

The table is best explained with a demonstrative example. If a blood culture is requested for a patient, then a blood sample will be taken and sent to the microbiology lab. The time at which this blood sample is taken is the `charttime`. The `spec_type_desc` will indicate that this is a blood sample. Bacteria will be cultured on the blood sample, and the remaining columns depend on the outcome of this growth:

- If no growth is found, the remaining columns will be NULL
- If bacteria is found, then each organism of bacteria will be present in `org_name`, resulting in multiple rows for the single specimen (i.e. multiple rows for the given `spec_type_desc`).
- If antibiotics are tested on a given bacterial organism, then each antibiotic tested will be present in the `ab_name` column (i.e. multiple rows for the given `org_name` associated with the given `spec_type_desc`). Antibiotic parameters and sensitivities are present in the remaining columns (`dilution_text`, `dilution_comparison`, `dilution_value`, `interpretation`).

Important considerations

Typically, negative values are indicated by a NULL value. However, `itemid` 90856 has a value of "NEGATIVE", and should be included in queries which seek to segregate microbiology data based on positive/negative findings.

`hadm_id` is assigned to observations using the administrative transfer table. However this does not always perfectly capture labs around the hospital stay. To be specific, as of v2.1, it is possible to assign 1,449,547 observations with an `hadm_id` using a join to *admissions* with `subject_id`, `admittime`, and `dischtime`. However, only 1,396,224 (96%) of these observations have an `hadm_id` actually stored in the *microbiologyevents* table. Users wishing to ensure capture of labs proximal to hospital stays should be aware of

this, and use joins with time as necessary.

Table columns

Name	Postgres data type	Example value
microevent_id	INTEGER NOT NULL	1234567
subject_id	INTEGER NOT NULL	12078372
hadm_id	INTEGER	29450599
micro_specimen_id	INTEGER NOT NULL	6386644
order_provider_id	VARCHAR(10)	P12ABC
chartdate	TIMESTAMP(0) NOT NULL	2130-04-01 00:00:00
charttime	TIMESTAMP(0)	2130-04-01 16:00:00
spec_itemid	INTEGER NOT NULL	70012
spec_type_desc	VARCHAR(100) NOT NULL	BLOOD CULTURE
test_seq	INTEGER NOT NULL	2
storedate	TIMESTAMP(0)	2130-04-05 00:00:00
storetime	TIMESTAMP(0)	2130-04-05 14:46:00
test_itemid	INTEGER	90117
test_name	VARCHAR(100)	ANAEROBIC BOTTLE
org_itemid	INTEGER	80155
org_name	VARCHAR(100)	STAPHYLOCOCCUS, COAGULASE NEGATIVE
isolate_num	SMALLINT	1

quantity	VARCHAR(50)	
ab_itemid	INTEGER	90025
ab_name	VARCHAR(30)	LEVOFLOXACIN
dilution_text	VARCHAR(10)	<=0.12
dilution_comparison	VARCHAR(20)	<=
dilution_value	DOUBLE PRECISION	0.12
interpretation	VARCHAR(5)	S
comments	TEXT	___

`microevent_id`

A unique integer denoting the row.

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`micro_specimen_id`

Uniquely denoted the specimen from which the microbiology measurement was made. Most microbiology measurements are made on patient derived samples (specimens) such as blood, urine, and so on. Often multiple measurements are made on the same sample. The `micro_specimen_id` will group measurements made on the same sample, e.g. organisms which grew from the same blood sample.

`order_provider_id`

`order_provider_id` provides an anonymous identifier for the provider who ordered the microbiology test. Provider identifiers follow a consistent pattern: the letter “P”, followed by either three numbers, followed by two letters or two numbers. For example,

"P003AB", "P00102", "P1248B", etc. Provider identifiers are randomly generated and do not have any inherent meaning aside from uniquely identifying the same provider across the database.

`chartdate`, `charttime`

`charttime` records the time at which an observation was charted, and is usually the closest proxy to the time the data was actually measured. `chartdate` is the same as `charttime`, except there is no time available.

`chartdate` was included as time information is not always available for microbiology measurements: in order to be clear about when this occurs, `charttime` is null, and `chartdate` contains the date of the measurement.

In the cases where both `charttime` and `chartdate` exists, `chartdate` is equal to a truncated version of `charttime` (i.e. `charttime` without the timing information). Not all observations have a `charttime`, but all observations have a `chartdate`.

`spec_itemid`, `spec_type_desc`

The specimen which is tested for bacterial growth. The specimen is a sample derived from a patient; e.g. blood, urine, sputum, etc.

`test_seq`

If multiple samples are drawn, the `test_seq` will delineate them. For example, if an aerobic and anerobic culture bottle are used for the same specimen, they will have distinct `test_seq` values (likely 1 and 2).

`storedate`, `storetime`

The date (`storedate`) or date and time (`storetime`) of when the microbiology result was available. While many interim results are made available during the process of assessing a microbiology culture, the times here are the time of the *last* known update.

`test_itemid`, `test_name`

The test performed on the given specimen.

`org_itemid`, `org_name`

The organism, if any, which grew when tested. If NULL, no organism grew (i.e. a negative culture).

`isolate_num`

For testing antibiotics, the isolated colony (integer; starts at 1).

`ab_itemid`, `ab_name`

If an antibiotic was tested against the given organism for sensitivity, the antibiotic is listed here.

`dilution_text`, `dilution_comparison`, `dilution_value`

Dilution values when testing antibiotic sensitivity.

`interpretation`

`interpretation` of the antibiotic sensitivity, and indicates the results of the test. "S" is sensitive, "R" is resistant, "I" is intermediate, and "P" is pending.

`comments`

Deidentified free-text comments associated with the microbiology measurement. Usually these provide information about the sample, whether any notifications were made to care providers regarding the results, considerations for interpretation, or in some cases the comments contain the result of the measurement itself. Comments which have been fully deidentified (i.e. no information content retained) are present as three underscores: `___`. A `NULL` comment indicates no comment was made for the row.

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MIMIC

MIMIC-IV / Hosp / omr table

omr

The Online Medical Record (OMR) table stores miscellaneous information documented in the electronic health record. It is a useful source of outpatient measurements such as blood pressure, weight, height, and body mass index.

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`chartdate`	DATE NOT NULL
`seq_num`	INTEGER NOT NULL
`result_name`	VARCHAR(100) NOT NULL
`result_value`	TEXT NOT NULL

Detailed Description

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`chartdate`

The date on which the observation was recorded.

`seq_num`

An monotonically increasing integer which uniquely distinguishes results of the same type recorded on the same day. For example, if two blood pressure measurements occur on the same day, `seq_num` orders them chronologically.

`result_name`

Each row provides detail regarding a single observation in the EHR. `result_name` provides a human interpretable description of the observation. As of MIMIC-IV v2.2, the following table lists the number of observations and the most common value.

result_name	number of observations	example value
eGFR	240	>60
Blood Pressure Lying	2764	120/70
Blood Pressure Sitting	3400	120/70
Blood Pressure Standing (1 min)	2560	90/60
Blood Pressure Standing	523	110/70
Blood Pressure Standing (3 mins)	626	80/50
BMI (kg/m2)	1662112	26.6
Weight (Lbs)	1889542	150
Blood Pressure	2169549	110/70
Height (Inches)	706906	64
BMI	554	25.0
Weight	354	150.00
Height	39	64.50

`result_value`

`result_value` is the value associated with the given OMR observation. For example, for

the `result_name` of 'Blood Pressure', the `field_value` column contains the recorded blood pressure (120/80, 130/70, and so on).

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MIMIC

MIMIC-IV / Hosp / patients table

Information that is consistent for the lifetime of a patient is stored in this table.

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`gender`	VARCHAR(1) NOT NULL
`anchor_age`	INTEGER NOT NULL
`anchor_year`	INTEGER NOT NULL
`anchor_year_group`	VARCHAR(255) NOT NULL
`dod`	TIMESTAMP(0)

Detailed Description

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual. As `subject_id` is the primary key for the table, it is unique for each row.

`gender`

`gender` is the genotypical sex of the patient.

`anchor_age`, `anchor_year`, `anchor_year_group`

These columns provide information regarding the actual patient year for the patient admission, and the patient's age at that time.

- `anchor_year` is a shifted year for the patient.
- `anchor_year_group` is a range of years - the patient's `anchor_year` occurred during this range.
- `anchor_age` is the patient's age in the `anchor_year`. If a patient's `anchor_age` is over 89 in the `anchor_year` then their `anchor_age` is set to 91, regardless of how old they actually were.
- Example: a patient has an `anchor_year` of 2153, `anchor_year_group` of 2008 - 2010, and an `anchor_age` of 60.
  - The year 2153 for the patient corresponds to 2008, 2009, or 2010.
  - The patient was 60 in the shifted year of 2153, i.e. they were 60 in 2008, 2009, or 2010.
  - A patient admission in 2154 will occur in 2009-2011, an admission in 2155 will occur in 2010-2012, and so on.

`dod`

The de-identified date of death for the patient. Date of death is extracted from two sources: the hospital information system and the Massachusetts State Registry of Vital Records and Statistics. Individual patient records from MIMIC were matched to the vital records using a custom algorithm based on identifiers including name, social security number, and date of birth.

As a result of the linkage, out of hospital mortality is available for MIMIC-IV patients up to **one year post-hospital discharge**. All patient deaths occurring more than one year after hospital discharge are censored. Survival studies should incorporate this into their design.

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MIMIC

pharmacy

The pharmacy table provides detailed information regarding filled medications which were prescribed to the patient. Pharmacy information includes the dose of the drug, the number of formulary doses, the frequency of dosing, the medication route, and the duration of the prescription.

Links to

- *poe* on `poe_id`
- *prescriptions* on `pharmacy_id`
- *emar* on `pharmacy_id`

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER NOT NULL
`pharmacy_id`	INTEGER NOT NULL
`poe_id`	VARCHAR(25)
`starttime`	TIMESTAMP(3)
`stoptime`	TIMESTAMP(3)
`medication`	TEXT
`proc_type`	VARCHAR(50) NOT NULL
`status`	VARCHAR(50)

entertime	TIMESTAMP(3) NOT NULL
verifiedtime	TIMESTAMP(3)
route	VARCHAR(50)
frequency	VARCHAR(50)
disp_sched	VARCHAR(255)
infusion_type	VARCHAR(15)
sliding_scale	VARCHAR(1)
lockout_interval	VARCHAR(50)
basal_rate	REAL
one_hr_max	VARCHAR(10)
doses_per_24_hrs	REAL
duration	REAL
duration_interval	VARCHAR(50)
expiration_value	INTEGER
expiration_unit	VARCHAR(50)
expirationdate	TIMESTAMP(3)
dispensation	VARCHAR(50)
fill_quantity	VARCHAR(50)

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`pharmacy_id`
A unique identifier for the given pharmacy entry. Each row of the pharmacy table has a unique `pharmacy_id`. This identifier can be used to link the pharmacy information to the provider order (in *poe* or *prescriptions*) or to the administration of the medication (in *emar*).

`poe_id`
A foreign key which links to the provider order entry order in the *prescriptions* table associated with this pharmacy record.

`starttime`, `stoptime`
The start and stop times for the given prescribed medication.

`medication`
The name of the medication provided.

`proc_type`
The type of order: “IV Piggyback”, “Non-formulary”, “Unit Dose”, and so on.

`status`
Whether the prescription is active, inactive, or discontinued.

`entertime`
The date and time at which the prescription was entered into the pharmacy system.

`verifiedtime`
The date and time at which the prescription was verified by a physician.

`route`
The intended route of administration for the prescription.

`frequency`
The frequency at which the medication should be administered to the patient. Many

commonly used short hands are used in the frequency column. Q# indicates every # hours; e.g. “Q6” or “Q6H” is every 6 hours.

`disp_sched`
The hours of the day at which the medication should be administered, e.g. “08, 20” would indicate the medication should be administered at 8:00 AM and 8:00 PM, respectively.

`infusion_type`
A coded letter describing the type of infusion: ‘B’, ‘C’, ‘N’, ‘N1’, ‘O’, or ‘R’.

`sliding_scale`
Indicates whether the medication should be given on a sliding scale: either ‘Y’ or ‘N’.

`lockout_interval`
The time the patient must wait until providing themselves with another dose; often used with patient controlled analgesia.

`basal_rate`
The rate at which the medication is given over 24 hours.

`one_hr_max`
The maximum dose that may be given in a single hour.

`doses_per_24_hrs`
The number of expected doses per 24 hours. Note that this column can be misleading for continuously infused medications as they are usually only “dosed” once per day, despite continuous administration.

`duration`, `duration_interval`
`duration` is the numeric duration of the given dose, while `duration_interval` can be considered as the unit of measurement for the given duration. For example, often `duration` is 1 and `duration_interval` is “Doses”. Alternatively, `duration` could be 8 and the `duration_interval` could be “Weeks”.

`expiration_value`, `expiration_unit`, `expirationdate`

If the drug has a relevant expiry date, these columns detail when this occurs. `expiration_value` and `expiration_unit` provide a length of time until the drug expires, e.g. 30 days, 72 hours, and so on. `expirationdate` provides the deidentified date of expiry.

`dispensation`

The source of dispensation for the medication.

`fill_quantity`

What proportion of the formulary to fill.

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MIMIC

MIMIC-IV / Hosp / poe

poe

Provider order entry (POE) is the general interface through which care providers at the hospital enter orders. Most treatments and procedures must be ordered via POE.

Links to

- *poe_detail* on `poe_id`

Table columns

Name	Postgres data type
`poe_id`	VARCHAR(25) NOT NULL
`poe_seq`	INTEGER NOT NULL
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER
`ordertime`	TIMESTAMP(0) NOT NULL
`order_type`	VARCHAR(25) NOT NULL
`order_subtype`	VARCHAR(50)
`transaction_type`	VARCHAR(15)
`discontinue_of_poe_id`	VARCHAR(25)
`discontinued_by_poe_id`	VARCHAR(25)
`order_provider_id`	VARCHAR(10)

`order_status`	VARCHAR(15)

`poe_id`

A unique identifier for the given order. `poe_id` is composed of `subject_id` and a monotonically increasing integer, `poe_seq`, in the following format: `subject_id`-`poe_seq`.

`poe_seq`

A monotonically increasing integer which chronologically sorts the POE orders. That is, POE orders can be ordered sequentially by `poe_seq`.

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`ordertime`

The date and time at which the provider order was made.

`order_type`

The type of provider order. One of the following:

- ADT orders
- Blood Bank
- Cardiology
- Consults
- Critical Care
- General Care
- Hemodialysis
- IV therapy
- Lab
- Medications
- Neurology

- Nutrition
- OB
- Radiology
- Respiratory
- TPN

`order_subtype`

Further detail on the type of order made by the provider. The `order_subtype` is best interpreted alongside the `order_type`, e.g. `order_type: 'Cardiology'` with `order_subtype: 'Holter Monitor'`.

`transaction_type`

The action which the provider performed when performing this order. One of the following:

- Change
- Co
- D/C
- H
- New
- T

`discontinue_of_poe_id`, `discontinued_by_poe_id`

If this order discontinues a previous order, then `discontinue_of_poe_id` will link to the previous order which was discontinued. Conversely, if this order was later discontinued by a distinct order, then `discontinued_by_poe_id` will link to that future order.

`order_provider_id`

`order_provider_id` provides an anonymous identifier for the provider who made the order. Provider identifiers follow a consistent pattern: the letter “P”, followed by either three numbers, followed by two letters or two numbers. For example, “P003AB”, “P00102”, “P1248B”, etc. Provider identifiers are randomly generated and do not have any inherent meaning aside from uniquely identifying the same provider across the database.

`order_status`

Whether the order is still active (‘Active’) or whether it has been inactivated (‘Inactive’).

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MIMIC

MIMIC-IV / Hosp / poe_detail

poe_detail

The *poe_detail* table provides further information on POE orders. The table uses an Entity-Attribute-Value (EAV) model: the entity is `poe_id`, the attribute is `field_name`, and the value is `field_value`. EAV tables allow for flexible description of entities when the attributes are heterogenous.

Links to

- *poe_detail* on `poe_id`

Table columns

Name	Postgres data type
`poe_id`	VARCHAR(25) NOT NULL
`poe_seq`	INTEGER NOT NULL
`subject_id`	INTEGER NOT NULL
`field_name`	VARCHAR(255) NOT NULL
`field_value`	TEXT

`poe_id`

A unique identifier for the given order. `poe_id` is composed of `subject_id` and a monotonically increasing integer, `poe_seq`, in the following format: `subject_id`-`poe_seq`.

`poe_seq`

A monotonically increasing integer which chronologically sorts the POE orders. That is, POE orders can be ordered sequentially by `poe_seq`.

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`field_name`

Each row provides detail regarding a particular aspect of a POE order. `field_name` is the name given to that aspect. As of MIMIC-IV v2.2, the following table lists the possible values and the most common entry for it in `field_value`.

`field_name`	number of rows	most frequent `field_value` observation
Admit to	881522	Medicine
Indication	27190	Antibiotics
Code status	197932	Resuscitate (Full code)
Transfer to	161707	Medicine
Admit category	1093726	Admit to inpatient
Consult Status	36591	Accepted
Discharge When	431642	Discharge Now
Level of Urgency	45617	Routine (within 24 hours)
Discharge Planning	475428	Finalized
Consult Status Time	36591	2149-12-03 09:00:01
Tubes & Drains type	491472	Indwelling urinary catheter (IUC) - Foley

`field_value`

`field_value` is the value associated with the given POE order and `field_name`. For example, for the `field_name` of 'Admit to', the `field_value` column contains the type of unit the patient was admitted to (Psychiatry, GYN, and so on).

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MIMIC

MIMIC-IV / Hosp / prescriptions

prescriptions

The *prescriptions* table provides information about prescribed medications. Information includes the name of the drug, coded identifiers including the Generic Sequence Number (GSN) and National Drug Code (NDC), the product strength, the formulary dose, and the route of administration.

Links to

- *pharmacy* on `pharmacy_id`
- *emar* on `pharmacy_id`

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER NOT NULL
`pharmacy_id`	INTEGER NOT NULL
`poe_id`	VARCHAR(25)
`poe_seq`	INTEGER
`order_provider_id`	VARCHAR(10)
`starttime`	TIMESTAMP(3)
`stoptime`	TIMESTAMP(3)
`drug_type`	VARCHAR(20) NOT NULL
`drug`	VARCHAR(255) NOT NULL

formulary_drug_cd	VARCHAR(50)
gsn	VARCHAR(255)
ndc	VARCHAR(25)
prod_strength	VARCHAR(255)
form_rx	VARCHAR(25)
dose_val_rx	VARCHAR(100)
dose_unit_rx	VARCHAR(50)
form_val_disp	VARCHAR(50)
form_unit_disp	VARCHAR(50)
doses_per_24_hrs	REAL
route	VARCHAR(50)

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`pharmacy_id`

An identifier which links administrations in *emar* to pharmacy information in the *pharmacy* table.

`poe_id`, `poe_seq`

These columns allow linking prescriptions to associated orders in the *poe* table.

`order_provider_id`

`order_provider_id` provides an anonymous identifier for the provider who initiated the

order. Provider identifiers follow a consistent pattern: the letter “P”, followed by either three numbers, followed by two letters or two numbers. For example, “P003AB”, “P00102”, “P1248B”, etc. Provider identifiers are randomly generated and do not have any inherent meaning aside from uniquely identifying the same provider across the database.

`starttime`, `stoptime`
The prescribed start and stop time for the medication.

`drug_type`
The component of the prescription which the drug occupies. Can be one of ‘MAIN’, ‘BASE’, or ‘ADDITIVE’.

`drug`
A free-text description of the medication administered.

`formulary_drug_cd`
A hospital specific ontology used to order drugs from the formulary.

`gsn`
The Generic Sequence Number (GSN), a coded identifier used for medications.

`ndc`
The National Drug Code (NDC), a coded identifier which uniquely identifiers medications.

`prod_strength`
A free-text description of the composition of the prescribed medication (e.g. ‘12 mg / 0.8 mL Oral Syringe’, ‘12.5mg Tablet’, etc).

`form_rx`
The container in which the formulary dose is delivered (e.g. ‘TABLET’, ‘VIAL’, etc).

`dose_val_rx`
The prescribed dose for the patient intended to be administered over the given time

period.

`dose_unit_rx`

The unit of measurement for the dose.

`form_val_disp`

The amount of the medication which is contained in a single formulary dose.

`form_unit_disp`

The unit of measurement used for the formulary dosage.

`doses_per_24_hrs`

The number of doses per 24 hours for which the medication is to be given. A daily dose would result in `doses_per_24_hrs`: 1, bidaily (BID) would be 2, and so on.

`route`

The route of administration for the medication.

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MIMIC

MIMIC-IV / Hosp / procedures_icd

procedures_icd

During routine hospital care, patients are billed by the *hospital* for procedures they undergo. This table contains a record of all procedures a patient was billed for during their hospital stay using the ICD-9 and ICD-10 ontologies.

Links to

- *d_icd_procedures* on `icd_code` and `icd_version`

Important considerations

- Procedures during the hospital stay can be billed (1) by the hospital or (2) by the provider. This table contains only procedures billed by the hospital.

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER NOT NULL
`seq_num`	INTEGER NOT NULL
`chartdate`	DATE NOT NULL
`icd_code`	VARCHAR(7)
`icd_version`	INTEGER

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows

associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`seq_num`

An assigned priority for procedures which occurred within the hospital stay.

`chartdate`

The date of the associated procedures. Date does *not* strictly correlate with `seq_num`.

`icd_code`, `icd_version`

`icd_code` is the International Coding Definitions (ICD) code.

There are two versions for this coding system: version 9 (ICD-9) and version 10 (ICD-10). These can be differentiated using the `icd_version` column. In general, ICD-10 codes are more detailed, though code mappings (or "cross-walks") exist which convert ICD-9 codes to ICD-10 codes.

Both ICD-9 and ICD-10 codes are often presented with a decimal. This decimal is not required for interpretation of an ICD code; i.e. the `icd_code` of '0010' is equivalent to '001.0'.

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MIMIC

MIMIC-IV / Hosp / provider table

provider

A description table for providers in the database referenced by *provider_id*. As of MIMIC-IV v2.2, this table simply lists all unique `provider_id` in the database.

Note that most tables contain a prefix before `provider_id`. All columns with a suffix link to `provider_id`, including:

- `admit_provider_id`
- `enter_provider_id`
- `order_provider_id`

Table columns

Name	Postgres data type
provider_id	VARCHAR(10) NOT NULL

`provider_id`

`provider_id` lists all possible identifiers for providers used throughout the database. Provider identifiers follow a consistent pattern: the letter “P”, followed by either three numbers, followed by two letters or two numbers. For example, “P003AB”, “P00102”, “P1248B”, etc. Provider identifiers are randomly generated and do not have any inherent meaning aside from uniquely identifying the same provider across the database.

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MIMIC

services

The *services* table describes the service that a patient was admitted under. While a patient can be physicially located at a given ICU type (say MICU), they are not necessarily being cared for by the team which staffs the MICU. This can happen due to a number of reasons, including bed shortage. The *services* table should be used if interested in identifying the type of service a patient is receiving in the hospital. For example, if interested in identifying surgical patients, the recommended method is searching for patients admitted under a surgical service.

Each service is listed in the table as an abbreviation - this is exactly how the data is stored in the hospital database. For user convenience, we have provided a description of each service type.

Service	Description
CMED	Cardiac Medical - for non-surgical cardiac related admissions
CSURG	Cardiac Surgery - for surgical cardiac admissions
DENT	Dental - for dental/jaw related admissions
ENT	Ear, nose, and throat - conditions primarily affecting these areas
EYE	Eye diseases - including subspecialty services in glaucoma, cataract surgery, cornea and external diseases, and neuro-ophthalmology.
GU	Genitourinary - reproductive organs/urinary system
GYN	Gynecological - female reproductive systems and breasts
MED	Medical - general service for internal medicine
NB	Newborn - infants born at the hospital
NBB	Newborn baby - infants born at the hospital

NMED	Neurologic Medical - non-surgical, relating to the brain
NSURG	Neurologic Surgical - surgical, relating to the brain
OBS	Obstetrics - conerned with childbirth and the care of women giving birth
ORTHO	Orthopaedic - surgical, relating to the musculoskeletal system
OMED	Oncologic Medical - non-surgical, relating to cancer
PSURG	Plastic - restortation/reconstruction of the human body (including cosmetic or aesthetic)
PSYCH	Psychiatric - mental disorders relating to mood, behaviour, cognition, or perceptions
SURG	Surgical - general surgical service not classified elsewhere
TRAUM	Trauma - injury or damage caused by physical harm from an external source
TSURG	Thoracic Surgical - surgery on the thorax, located between the neck and the abdomen
VSURG	Vascular Surgical - surgery relating to the circulatory system

Links to

- patients on `subject_id`
- admissions on `hadm_id`

Table columns

Name	Postgres data type
`subject_id`	INT
`hadm_id`	INT
`transfertime`	TIMESTAMP(0)

`prev_service`	VARCHAR(20)
`curr_service`	VARCHAR(20)

`subject_id`

`subject_id` is a unique identifier which specifies an individual patient. Any rows associated with a single `subject_id` pertain to the same individual.

`hadm_id`

`hadm_id` is an integer identifier which is unique for each patient hospitalization.

`transfertime`

`transfertime` is the time at which the patient moved from the `prev_service` (if present) to the `curr_service`.

`prev_service`, `curr_service`

`prev_service` and `curr_service` are the previous and current service that the patient resides under.

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MIMIC

MIMIC-IV / Hosp / transfers table

transfers

Physical locations for patients throughout their hospital stay.

Links to

- *patients* on `subject_id`
- *admissions* on `hadm_id`

Important considerations

- The `icustays` table is derived from this table.

Table columns

Name	Postgres data type
`subject_id`	INTEGER NOT NULL
`hadm_id`	INTEGER
`transfer_id`	INTEGER NOT NULL
`eventtype`	VARCHAR(10)
`careunit`	VARCHAR(255)
`intime`	TIMESTAMP(0)
`outtime`	TIMESTAMP(0)

Detailed Description

`subject_id`, `hadm_id`, `transfer_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay, and `transfer_id` is unique to a patient physical location.

Note that `stay_id` present in the *icustays* and *edstays* tables is derived from `transfer_id`. For example, three contiguous ICU stays will have three separate `transfer_id` for each distinct physical location (e.g. a patient could move from one bed to another). The entire stay will have a single `stay_id`, whih will be equal to the `transfer_id` of the first physical location.

`eventtype`

`eventtype` describes what transfer event occurred: 'ed' for an emergency department stay, 'admit' for an admission to the hospital, 'transfer' for an intra-hospital transfer and 'discharge' for a discharge from the hospital.

`careunit`

The type of unit or ward in which the patient is physically located. Examples of care units include medical ICUs, surgical ICUs, medical wards, new baby nurseries, and so on.

`intime`, `outtime`

`intime` provides the date and time the patient was transferred into the current care unit (`careunit`) from the previous care unit. `outtime` provides the date and time the patient was transferred out of the current physical location.

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